

**THE RELATIONSHIP BETWEEN DYSCALCULIA AND ANXIETY
LEVELS IN UNIVERSITY STUDENTS IN NIGERIA: A CROSS-
SECTIONAL ANALYSIS**

BY

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CERTIFICATION

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DEDICATION

I dedicated this work to the Almighty God for His unmerited favor during the purpose of the study.

ACKNOWLEDGEMENT

I give glory and adoration to God almighty for making this research work a success.

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ABSTRACT

Dyscalculia, a specific learning disability in mathematics, has increasingly become a concern for university students, as it can significantly impact their academic success and emotional well-being. Concurrently, anxiety levels, including the specific variant known as mathematics anxiety, have been on the rise among students in higher education. However, a notable gap exists in the literature concerning the intricate relationship between dyscalculia and anxiety within this demographic.

This research project embarked on a cross-sectional analysis with the aim of delving into the potential correlation between dyscalculia and anxiety levels in a diverse sample of university students. The overarching objective of this study was to provide a deeper understanding of the academic and psychosocial challenges faced by students grappling with dyscalculia, while also seeking to offer insights into possible interventions that could support their educational journeys. In the pursuit of these goals, students from different universities and departments were used for this research. Data was collected using the Dyscalculia Assessment Questionnaire (DAQ), to test for dyscalculia in students and the Math Anxiety Assessment Questionnaire (MAAQ). Subsequently, the amassed data was meticulously analyzed through a suite of statistical techniques, including correlation analysis.

The projected outcomes of this research held the promise of unveiling key insights into the potential correlation between dyscalculia and anxiety levels among university students. Through this inquiry, we anticipated gaining a clearer understanding of the psychosocial ramifications of dyscalculia, its multifaceted impact on academic performance, and the intricate interplay between cognitive challenges and emotional well-being.

In conclusion, this research endeavor, addressing the long-standing research gap concerning the dynamic relationship between dyscalculia and anxiety within the university student population, contributed vital information. This information, benefits educators, psychologists, and institutions alike, is expected to inform the development of targeted interventions and support mechanisms. Ultimately, outcome is an enhanced academic experience and improved well-being for all university students, particularly those grappling with the challenges of dyscalculia.

CHAPTER ONE

INTRODUCTION

Background of the Study

Mathematics is perceived to be a difficult subject due to its abstract nature. Being a very important and necessary subject in school education due to its effect on everyday life and human activities, it is usually given special attention in school education globally. According to Evans (2000), science is the bedrock that provides the springboard for the growth of technology and mathematics is the gate and key to sciences. Math knowledge is positively linked with prosperity in our industry, technological society (Peri et al, 2016)

Many students believe that it takes only brilliance to perform well rather than critical reading, determination and perseverance. The expected outcomes of teaching mathematics could not be achieved and students' negative attitude towards the subject could not be changed as there are still ongoing research on the phobia of students for mathematics and possible solutions have been given which is seen to be less effective. Mathematics stays at the edge of extinction because new research shows that students' interest in mathematics is declining.

For many years, it was believed that the numerical cognition of learners could be developed effectively using Piaget's child developmental stages.

But however, in recent times, researchers are focusing on causes of mathematical learning difficulties in learners as well as neurological factors. Researchers have generally agreed that the deficit in working memory, brain related condition, genetic cause and environmental and brain difference is called dyscalculia. Dyscalculia is seen to affect not just children, but also adults.

Approximately 3 to 6% of children are affected from special disorders of numerical understanding (developmental dyscalculia (DD)). Impaired development of number processing skills in these children is characterized by problems in various aspects of numeracy as well as alterations of brain activation and brain structure (Karin and Michael, 2015).

Over the years, little effort has been put into discovering causes and solutions to dyscalculia with its effect leading to Attention Deficit Hyperactivity Disorder (ADHD). Due to the uprising of the case in Nigeria, there's a need for a critical investigation and study of the developmental dyscalculia, its causes and effects on learning process.

Dyscalculia shares a close relationship with Math anxiety in the academic performance of students. It has been seen that students' fear for mathematics is a symptom of dyscalculia but math anxiety does not necessarily mean it's a case of dyscalculia.

Dyscalculia has been defined by many researchers and neurosurgeons in quite a few ways. Generally, dyscalculia can be defined as a disability that results in difficulty leaning or comprehending arithmetic, such as difficulty in understanding numbers, learning how to manipulate numbers, performing mathematics calculations, and learning facts in mathematics.

The term 'dyscalculia' was coined in 1940's by Dr. Gerstmann. He was the first to write a paper on a rare neurological disorder that can occur as a result of brain injury or as a development disorder. He coined the term from Greek and Latin words. The Greek prefix 'dys' which means 'badly' and calculia from the Latin word 'calulare' which means 'to count'.

Dyscalculia was first discovered in 1919 by a Swedish Neurologist Solomon Henschen. He discovered that it was possible for a person of high general intelligence to have impaired mathematics abilities. At that time, it was called 'number blindness'.

The developmental dyscalculia was first seen by a Czechoslovakian Researcher Dr. Vladislav Kosc in 1974. In his article *Developmental Dyscalculia (DD)*, he defined developmental dyscalculia as ‘a structural disorder of mathematical abilities which has its origin in a genetic or congenital disorder in those parts of the brain that are the anatomical psychological substrate of the maturation of the mathematical abilities adequate to age, without a simultaneous disorder of general mental functions’. Dr. Kosc originally separated dyscalculia into six (6) categories based on students’ abilities or tasks that they were having trouble with. They include verbal dyscalculia, practognostic dyscalculia, lexical dyscalculia, graphical dyscalculia, ideognostical dyscalculia, and operational dyscalculia.

The DSM-IV (2000) document, used by educational psychologists, defines mathematics disorder in terms of test scores: as measured by a standardized test that is given individually, the person’s mathematical ability is substantially less than would be expected from the person’s age, intelligence and education. This deficiency materially impedes academic achievement or daily living’.

The National Numeracy strategy (DfES, 2001) gives the following definitions: ‘Dyscalculia is a condition that affects the ability to acquire arithmetical skills. Dyscalculic learners may have difficulty understanding simple number concepts, lack an intuitive grasp of numbers, and have problems learning number facts and procedures. Even if they produce a correct answer or use the correct method, they may do so mechanically and without confidence’.

Anxiety level refers to the degree of anxiety or the intensity of the emotional response an individual experiences in a given situation. Anxiety is a natural and common emotional response that helps us prepare for potential threats and challenges. But when it becomes excessive, persistent, or interferes with daily functioning, it may indicate an anxiety disorder.

Factors affecting anxiety level may include stressors, coping mechanisms, personality, environmental factors, past experiences; etc.

Math anxiety is defined by feelings of tension, helplessness, mental disorganization, and associated bodily symptoms that are evoked in mathematical problem-solving situations (Ashcraft, 2002).

The overview of mathematics anxiety is rooted in the accounts of what was described as “mathemaphobia” (Gough, 1954), or “number anxiety” (Dreger & aiken, 1957).

Math Anxiety as defined by Richardson and Suinn (1972) in the Mathematics Anxiety Research Scale (MARS) is the feeling of tension and anxiety that disrupts the manipulation of numbers and the solving of mathematical problems.

According to the Deficit Theory, it suggests that people who begin with poorer results in mathematics are more likely to develop math anxiety.

Often times, Dyscalculia and Math Anxiety are misinterpreted as both seem to look same when dealing with reason for mathematics failure. When a child fails mathematics, it is believed that it is because the child fears mathematics then becomes anxious when given a mathematical problem. Only a few people would try to test the child for other factors that might have influenced the learning of mathematics.

Up to the tertiary level, most students study courses they know they need no knowledge of mathematics because of the fear of mathematics or the fact that they don’t know it no matter how many times they study it. And when they face it, they tend to flop and face issues.

Math anxiety (MA) has been seen to decrease the mathematics performance of students. Studies that explored the math anxiety-performance link, conducted from 2000 to 2019 (84 samples,

N=8680), were identified and statistically integrated with a meta-analysis method. The results indicated a robust negative math anxiety-performance link.

Statement of Problem

Dyscalculia, a specific learning disorder affecting mathematics comprehension and skills, has been seen as a challenge for many individuals, which includes most university students. In the same vein, anxiety levels among university students have become a concern due to their potential impact on academic performance, mental well-being, and overall quality of life. Despite extensive study on the influence of dyscalculia on academic performance, cognitive development, and potential psychological consequences, there is a need to determine whether there is a significant association between dyscalculia and anxiety in the setting of university students. With a specific focus on university students, this study seeks to close this knowledge gap and advance understanding of the interactions between learning disorders and mental health issues.

Research Questions

This study will solely focus on finding answers to:

1. Is there a statistically significant correlation between the presence of dyscalculia and anxiety levels among university students?
2. To what extent do different aspects of dyscalculia such as numerical processing, mathematical reasoning, and arithmetic skills correlate with varying levels of anxiety in University students?
3. Are there any demographic factors such as age, gender, and field of study that moderate the relationship between dyscalculia and anxiety among university students?

Research Hypothesis

H1: There is a significant correlation between the presence of dyscalculia and anxiety levels among university students.

H2: Different aspects such as numerical processing, mathematical reasoning, and arithmetic skills, correlate with varying levels of anxiety in University students.

H3: Demographic factors such as age, gender, and field of study, moderate the relationship between dyscalculia and anxiety among university students.

Significance of the Study

This study is considered to be very important because it's aimed at analyzing and developing learners with difficulty in mathematics due to dyscalculia and mathematics anxiety

In addition, this study is also significant because its findings have implications for:

1. Curriculum planners and experts.
2. The parents.
3. The lecturers.
4. School administrators.
5. The guidance counselors, among others.

For example, the result will help the curriculum planners know the impact of dyscalculia and mathematics anxiety in learning process and develop a curriculum suitable for all learners.

Scope and Delimitation

Conceptually, the study will include dyscalculia status as the independent variable and anxiety levels as the dependent variable. The geographical scope of this study is the Nigeria

Operational Definition of Terms

1. Dyscalculia: Dyscalculia is a learning disorder that affects a person's ability to understand number-based information and mathematics.
2. Mathematics anxiety: This is a psychological condition marked with feelings of worry or tension when confronted with mathematical tasks or situations.

CHAPTER 2

LITERATURE REVIEW

CONCEPTUAL FRAMEWORK

DYSCALCULIA

Dyscalculia, also known as “math dyslexia”, is a neurological condition that affects an individual’s ability to understand and process mathematical concepts, leading to difficulties in numerical mathematics.

Dyscalculia can be defined as a disability that result in difficulty leaning or comprehending arithmetic, such as difficulty in understanding numbers, learning how to manipulate numbers, performing mathematics calculations, and learning facts in mathematics.

The National Numeracy strategy (DfES, 2001) gives the following definitions: ‘Dyscalculia is a condition that affects the ability to acquire arithmetical skills. Dyscalculic learners may have difficulty understanding simple number concepts, lack an intuitive grasp of numbers, and have problems learning number facts and procedures. Even if they produce a correct answer or use a correct method, they may do so mechanically and without confidence’.

The DSM-IV (2000) document, used by educational psychologists, defines mathematics disorder in terms of test scores: ‘as measured by a standardized test that is given individually, the person’s mathematical ability is substantially less than would be expected from the person’s age, intelligence and education. This deficiency materially impedes academic achievement or daily living’.

The American Psychiatric Association (2013) defines Developmental Dyscalculia as a specific learning disorder that is characterized by impairment in learning basic arithmetic facts, processing numerical magnitude and performing accurate and fluent calculations.

Generally, it is agreed that around 5-8% of learners suffer from dyscalculia and on average, each class of 30 learners will have approximately two or three persons who are affected by it.

Research in Washington in 2018 showed that nearly 48,000 children were identified as having a specific learning disability (SLD) which includes dyscalculia, dyslexia, and dysgraphia.

It has been observed by Yusha'u (2006) that difficulties at school that are related to dyscalculia have been the subject of less research in Sokoto. Thereby results in stigmatization of learners.

'Number sense' issues are the most prevalent. This is a natural grasp of how numbers function, as well as how to compare and gauge quantities on a number line. This means that children with dyscalculia have trouble understanding number words (one, two, twelve, twenty), numerals (one, two, twelve, twenty), and their relationships (i.e., which quantity is greater or less than the other, etc.), as well as the number line and the reasoning behind its sequential order. These problems progress into being unable to perform simple mental calculations, apply the four mathematical operations of addition, subtraction, multiplication, and division, struggle with "missing operand problems" ($13 + ? = 46$), struggle with word problems, and eventually struggle with time and money management, etc.

Studies show that even babies have a basic sense of numbers. Dr. Brian Butterworth, a leading researcher in dyscalculia, compares number sense to being color-blind. He says some people are born with number blindness. This makes it hard to tell the difference between quantities. Number

blindness is one reason many kids have trouble connecting numbers to the real world. They can't grasp the idea that 'five cookies' has the same number of objects as 'five cakes' and 'five apples'.



Despite several researches carried out on dyscalculia, the fact still remains that there are still students in and out of schools with difficulties when it comes to learning mathematics. This can be noted when a child is yawning for an English class and will not want to attend a mathematics class. Other indications of dyscalculia among students are performing below expectations, discrepancy between developmental level and cognitive ability, laterality, rote counting, rules and formulae, sequencing, time management, etc.

One of the challenges in identifying the behavioral profile of DD is that math difficulties can appear at different times of the educational process.

Mazzocco and Myers (2003), for example, showed that 65% of a sample of DD children in grade 3 had fulfilled the DD diagnosis criteria in kindergarten, whereas 20% of the sample had met the criterion for the first time in grade 2. This discovery emphasizes the fact that the abilities required for successful mathematics performance fluctuate during development, and so certain children

may have a specific impairment at an early learning stage, disrupting the acquisition of later skills. This disruption may develop as a result of a lack of core abilities. This extra effort put into basic tasks makes children unable to follow and learn.

In the classroom, more complicated procedural information is taught (Pellegrino and Goldman 1987). The discovery of changing profiles over time emphasizes the significance of looking for persistent deficiencies in the diagnosis of DD. In other words, it is critical that children have deficiencies in math skill at more than one point in time.

History of Dyscalculia

The term ‘dyscalculia’ was coined in 1940’s by Dr. Gerstmann. He was the first to write a paper on a rare neurological disorder that can occur as a result of a brain injury or as a development disorder. He coined the term from Greek and Latin words. The Greek prefix ‘dys’ which means ‘badly’ and calculia from the Latin word ‘calulare’ which means ‘to count’.

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practognostic dyscalculia, lexical dyscalculia, graphical dyscalculia, ideognostical dyscalculia, and operational dyscalculia.

In the late 1990s, almost 40 years later, as the use of MRI equipment for diagnosing brain problems increased, further study was done. According to the findings, dyscalculia is a condition in which one or more brain regions that are in charge of doing simple calculations are unable to coordinate properly.

David Geary divided his research into three categories—semantic, procedural, and visual-spatial memory—to further examine the reasons of the dyscalculia. All pupils who scored below the 35th percentile on the Woodcock-Johnson Mathematics thinking test were considered to be students with mathematical impairments, according to Geary. To describe the dyscalculia, he referred to it as a "Math disability."

Categories of Dyscalculia

According to Dr. Kosci, dyscalculia can be divided into six (6) categories. They include;

Verbal Dyscalculia: The hallmark of verbal dyscalculia is a challenge naming and comprehending the mathematical ideas spoken aloud. People with this form of dyscalculia can read and write numbers, but they struggle to recognize them when they are spoken. Communication in language and mathematics is the focus of verbal dyscalculia. This type of dyscalculia makes it difficult for sufferers to understand and use mathematical words, symbols, and language. They might have trouble reading and comprehending texts, word puzzles, and instructions that deal with math. It might be challenging to convert from mathematical equations to real-world situations and vice versa. As the linguistic component is essential for engaging with mathematical concepts, this category might make it difficult to understand basic mathematical processes.

1. **Operational Dyscalculia:** Mathematical calculations and procedures are examined in procedural dyscalculia. People who fall into this category have trouble performing mathematical procedures. Mathematical operations like addition, subtraction, multiplication, and division can be quite difficult. The sequential procedures needed to answer arithmetic problems could get complicated and result in computation errors. For one to advance to more complicated mathematical concepts, one must have a firm grasp of mathematical techniques.
2. **Semantic Dyscalculia:** The conceptual comprehension of numbers and mathematical concepts is relevant to semantic dyscalculia. This type of dyscalculia makes it difficult for sufferers to understand the connections between numbers and their magnitudes. Place value and number sense may continue to be mysterious concepts. It can be particularly challenging to estimate quantities, compare numbers, and understand abstract mathematical concepts. Building advanced mathematical skills requires a strong basis in semantic understanding.
3. **Graphical Dyscalculia:** Quantitative visualization and spatial interactions are central to visual-spatial dyscalculia. People who encounter this difficulty have trouble understanding spatial layouts, numerical orientation, and geometric shapes. It can be difficult to understand graphs, charts, and maps, and geometry-related issues can seem impossible to solve. Understanding advanced geometry and applied mathematics requires the capacity to visualize and manipulate spatial information.
4. **Ideognostical Dyscalculia:** Working memory dyscalculia exposes flaws in the working memory system that interferes with manipulating and remembering numerical information. People in this category have trouble retaining and processing numerical information for brief periods of time. It can be particularly challenging to follow directions that need multiple steps,

to calculate with intermediate results, and to mentally manipulate numbers. For effectively handling mathematical tasks and processes, a strong working memory is required.

5. **Practognostic Dyscalculia:** The focus of practognostic dyscalculia is on tactile and practical estimation. Individuals suffering from this type of dyscalculia have difficulty judging quantities through touch and manipulating things in numerical contexts. Simple tasks requiring practical estimating and measuring may be challenging. The ability to estimate quantities through interacting with the physical world is critical in a variety of real-life settings.

Characteristics of Dyscalculia

Dyscalculia is a phrase that has gained popularity in recent years. However, there is still considerable disagreement concerning the nature of the illness, and numerous definitions are in use (Finesilver & Rodd, 2017). Most teachers are unaware of the characteristics, symptoms, and intervention techniques for dyscalculia (Sousa, Dias, & Cadime, 2016). Dyscalculic students are uninterested in mathematics. They have difficulty deciding which arithmetic operation to use: divide, multiply, add, or subtract. They also struggle to determine which side is left and which is right. Little challenges that we face in our daily lives have a huge impact on the individual, socially and psychologically. As a result, individuals may acquire a math phobia (Ferraz & Neves, 2015).

Dyscalculia is a specific learning disorder with impairment in mathematics in the areas of number sense, memorization of arithmetic facts, accurate or fluent calculation, and accurate math reasoning, according to the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) (American Psychiatric Association, 2013). In short, a dyscalculic student has some limitations or difficulties understanding arithmetic when compared to their counterparts of the same age. Thus in this paper, the researcher will discuss the characteristics of dyscalculia based on four domains,

namely; (1) number sense; (2) working memory; (3) accurate of fluent calculation; and (4) mathematics reasoning.

1. Number Sense: Number sense is the ability to process estimated numbers spontaneously (Hannagan, Nieder, Viswanathan, & Dehaene, 2017). Number sense is required for subitizing, approximate calculation, comparison, bisection, and computations such as subtraction that cannot be handled with basic arithmetic facts alone (Karakonstantaki, Simos, Michalis, & Micheloyannis, 2017). It has been discovered to be a fundamental numerical aptitude for youngsters learning early mathematics (Chinn & Ashcroft, 2016). Number sense is essential since it is related to mathematical ability and helps explain why certain youngsters struggle with math. Youngsters who lack number sense will have worse arithmetic skills than youngsters who do not lack number sense. Three tasks can be used to assess it: dot comparison, number comparison, and number line 1-100 (Kroesbergen & van Dijk, 2015).

Overall, number sense is a natural skill that everyone is born with. It is the foundation for all students to master numeracy abilities such as numbers and arithmetic. Dyscalculic students have a poor number sense, resulting in poor arithmetic skills in comparison to their peers. As a result, number sense can be one of the most crucial indications of dyscalculia in students.

2. Working Memory: When investigating short-term storage capacity in developmental dyscalculia, working memory for serial order may be a significant component to explore (Attout & Majerus, 2015). Dyscalculia is caused by a lack of working memory. This makes memorizing math knowledge and applying them to a math issue more difficult (Alloway & Alloway, 2015). Some students were unable to retain multiplication facts due to differences in working memory levels (Yoong & Noor Aini Ahmad, 2018).

When students' limited working memory is directed to superfluous or duplicate material, they do not learn well (Centre for Education Statistics and Evaluation, 2017). Working memory, in a nutshell, is a restricted capacity that is responsible for temporarily storing information. One of the predictors of dyscalculia is short-term and long-term memory. Students who have problems with working memory also have problems in math. As a result, limited working memory can be used to screen for dyscalculic students.

3. **Accurate and Fluent Calculation:** The cornerstone of word problems is calculation. Calculation problems are influenced by difficulty in numerical operations vocabulary and procedural expertise (Peng & Lin, 2019). A substantial body of work indicates the roles and contributions of calculation accuracy and calculation fluency in a wide range of mathematical outcomes (Fuchs et al., 2016). Early primary grade computation accuracy and fluency can be utilized to predict later mathematical achievement (Rittle-Johnson, Fyfe, Hofer, & Farran, 2016).

Calculation, in short, involves the four basic arithmetic operations of addition, subtraction, multiplication, and division. Accurate calculation refers to a student's ability to perform arithmetic operations accurately and without errors, whereas fluent calculation refers to a student's ability to answer arithmetic operations quickly and without pausing to comprehend each and every method. Inability to calculate accurately or fluently is thus another indicator for recognizing dyscalculic students.

4. **Mathematics Reasoning:** Mathematics is rich enough to allow for various routes to legitimate mathematical reasoning (Goldstone, Marghetis, Weitnauer, Ottmar, & Landy, 2017). Mathematics reasoning is defined as a process of communication with others or with oneself that allows for the inference of mathematical utterances from other mathematical utterances (Jeannotte & Kieran,

2017). As a result, the ability to reason mathematically encompasses the ability to formulate questions, solve problems, and justify solutions (Mata-Pereira & da Ponte, 2017).

In this study, mathematic reasoning is defined as the capacity to grasp mathematical questions, solve problems, and determine if a solution is correct or incorrect by providing reasonable rationale.

Discussion

The researcher summarized 20 characteristics of dyscalculia based on the four domains as described in the literature review. Subitizing, sequencing, counting backwards, number recognition, and poor money abilities are all features of dyscalculia in the domain of number sense (Alloway & Alloway, 2015; Chinn, 2016; Laurillard, 2016; Bird, 2017). Furthermore, short-term memory, long-term memory, getting lost in multi-step word problems, struggling with mental arithmetic, struggling with multiplication facts, and forgetting mathematical procedures are characteristics of dyscalculia under the domain of working memory (Alloway & Alloway, 2015; Chinn, 2016; Bird, 2017).

As a result, the symptoms of dyscalculia in the domain of accurate or fluent calculation include difficulty learning to count, slow processing speed, counting on for addition facts, difficult to count sequences, and using tally marks for addition or subtraction (Alloway & Alloway, 2015; Chinn, 2016; Bird, 2017). Last but not least, the hallmarks of dyscalculia in the realm of mathematics reasoning include estimating, difficulties progressing, difficulty assessing, and using formulae without comprehending (Chinn, 2016). The characteristics of dyscalculia are shown in the table below.

Domain	Characteristics of Dyscalculia
---------------	---------------------------------------

1. Number Sense	• Subitizing • Sequencing • Counting Backwards • Recognising Numbers • Poor skills with money
2. Working Memory	• Short-term Memory • Long-term Memory • Gets lost in multi-step word problem • Struggle with mental arithmetic • Struggle with multiplication facts • Forgets mathematical procedures
3. Accurate and Fluent Calculation	• Trouble in learning to count • Slow processing speeds • Counts on for addition facts • Difficult to count sequences • Use tally marks for addition or subtraction
4. Mathematics Reasoning	• Estimating • Difficulty in progressing • Difficulty in judging • Use formulas without understanding

Characteristics of Dyscalculia in Learning Mathematics

Mathematics Anxiety

Anxiety disorders affect over 40 million people in the United States, according to the Anxiety and Depression Association of America (ADAA). It is the most frequent type of mental disease in the United States. However, only 36.9% of those suffering from an anxiety problem obtain therapy.

Anxiety is defined by the American Psychological Association (APA) as "an emotion characterized by feelings of tension, worried thoughts, and physical changes like increased blood pressure."

Anxiety is a complex reaction to real or perceived threats. It can include cognitive, physical, and behavioral changes.

Adrenaline, a hormone and chemical messenger in the brain, is released in response to real or imagined danger, triggering these anxiety emotions in a process known as the fight-or-flight response. Some people may have this reaction in tough social situations or when confronted with major events or decisions.

Maths anxiety is characterized as a sensation of tension, uncertainty, and anxiety that interferes with math performance ability, number manipulation, and problem solving in a wide range of everyday life and academic contexts. Maths anxiety, according to Olango, is an affective, behavioral, and cognitive response to a perceived danger to self-esteem that develops in response to mathematical problems. In two ways, math anxiety, which is rooted in emotional reasons, can be distinguished from dyscalculia, which is defined by a specific cognitive weakness in mathematics. For starters, math anxiety might arise in persons who have arithmetic talent but despise math.

Green and Allerton (1999) defined mathematics anxiety as a state of mind formed from personal experience and an individual's emotional reaction to these events. They went on to say that

unfavorable attitudes about learning mathematics occur as a result of a variety of contacts with how mathematics is presented, taught, and learned by individuals.

According to Sheffield and Hunt (2007), mathematics anxiety is the feeling of anxiety that certain people experience when confronted with mathematical issues. They also claim that, like other types of anxiety, students may feel their heartbeat faster or stronger, believe they are incapable of solving mathematical problems, and avoid taking mathematics classes.

Mathematics anxiety has also been found to correlate with measures of general anxiety, which may serve as a background variable explaining some of the association between mathematics anxiety and test anxiety. Hembree (1990), for example, discovered a 0.35 mean association between the MARS and a measure of overall anxiety. Wang et al. (2014) found evidence that genetic variation in general anxiety lead to genetic differences in mathematics anxiety in a behavioral genetic investigation.

Mathematics anxiety cannot be reduced to either exam anxiety or general anxiety. Different measures of mathematics anxiety have a higher correlation (0.5-0.8) than exam anxiety or general anxiety (Dew et al., 1983; Hembree, 1990; review by Ashcraft and Ridley, 2005).

Math anxiety may also impair academic performance by taxing working memory (Ashcraft, 2016; Skagerlund et al., 2019; Pelegrina et al., 2020). Many research have indicated that anxiety, in general, is connected with working memory deficiencies (Caviola et al., 2017; Eysenck and Calvo, 1992). As a result, the burden that math anxiety and related reflections place on working memory may explain the deterioration in arithmetic performance (Engle, 2016; Maloney et al., 2013).

Math anxiety can arise at any level of schooling, from elementary school to university. In a sample of first-grade pupils, Harari et al. discovered that negative responses and numerical confidence are

the most prominent characteristics of math fear. Similar findings were made at the postsecondary level in a variety of subjects, including health care professions. Roykenes and Larsen, for example, evaluated 116 baccalaureate nursing students and discovered a negative link between past mathematic likes/dislikes and self-assessment of mathematic aptitude.

Many factors can lead to or alleviate math anxiety. Teachers, parents, classmates, and society may all be factors or facilitators. Arithmetic anxiety can develop as a result of negative arithmetic learning experiences in the classroom or at home. To begin with, the teacher plays a vital role in making the lesson more appealing and lowering worries. Good math teachers can foster a learning atmosphere in which students have high expectations for their progress. Second, parents have a crucial role in growing or lessening their children's math fear. The actions and relationships of parents with their children are crucial in this regard. Parents can identify any learning problems in their children at an early stage by sharing their anxieties and phobias.

This may prevent the students from getting any learning worries later in life. Furthermore, parents' math anxiety causes their children to learn fewer maths during the school year and to experience more math anxiety by the conclusion of the school year. Third, peers play an essential role in reducing math anxiety. Peers at any stage of learning can have a detrimental impact on their colleagues, for example, when students commit blunders in front of their peers. Finally, due to misconceptions about mathematics, or math myths, society might contribute to the development of math anxiety.

Determinants of Math anxiety

Mathematical anxiety is an issue that is influenced by both internal and external influences. Internal influences include genes that influence nervous system development, as well as external environmental and socio-cultural factors.

Internal Elements that Influence Math Anxiety

Internal factors that contribute to arithmetic anxiety include neurological issues, heredity, and age.

1. Math Anxiety and Neurological Variables

Math anxiety is linked to neurobiological variables. The following mathematical skills determine mathematical reasoning ability:

- 1) Ability to answer elementary arithmetic problems,
- 2) Ability to compare values and symbols,
- 3) Ability to compute verbally, and
- 4) Ability to tackle more complicated mathematical problems (geometry and spatial synthesis).

Currently, mathematical abilities are thought to be gained mostly through the activity of the hippocampus, as well as the parietal and temporal lobes. Some research (Jolles et al., 2016; Supekar et al., 2015; Uddin et al., 2010) found that mathematical talents in children are connected with the size of the hippocampus and that mathematical abilities in adults are correlated with the size of the posterior and temporal-parietal lobes. It has also been observed that children with math anxiety had amygdala and medial prefrontal cortex activation as well as decreased interparietal sulcus activity (Qin et al., 2014).

2. Math Anxiety and Genes

It has been proposed that hereditary variables are linked to mathematical cognition and anxiety in general. The results of twin research suggest that monozygotic twins have a concordance of 70% and dizygotic twins have a concordance of 50% (Hart et al., 2016; Wang et al., 2014). Mathematical anxiety has been linked to changes in gene groups involved in the expression of neuronal proteins, neurotransmitters, working memory, and synaptic plasticity. Working memory genes contribute to the formation of prefrontal cortex excitation and inhibition systems, which comprise glutamatergic pyramidal neurons and gabaergic inhibitory inter neurons. Mathematical anxiety has also been linked to changes in the expression of genes involved in synaptic plasticity.

3. Math anxiety and Age

Math anxiety appears to rise with age in general, with studies indicating that severe math anxiety is uncommon in children (Barroso et al., 2021; Szczygiel, 2020; Ashcraft, 1992; Carey et al., 2016; Gough, 1954; Hembree, 1990). The emergence of clinical anxiety disorders peaks in early adolescence, and general anxiety increases with age (Barroso et al., 2021; Szczygiel, 2020; Egger and Angold, 2006; Kilts et al., 2006). Arithmetic with larger numbers, which places a greater demand on working memory, and more abstract numerical parts of mathematics, might raise anxiety in adolescents.

External Factors that Influence Math Anxiety

Gender, nationality, and environment are all external factors that contribute to math fear.

1. Math Anxiety and Gender

According to research, there is little or no difference in math achievement between men and women in countries with educational fairness. They do, however, show that women grade themselves lower and have higher anxiety about mathematics (Beilock et al., 2007; Baird and

Keene, 2019; Barroso et al., 2021). The threat of stereotype is one possible explanation for more math anxiety in women than in males (Beilock et al., 2007; Buratta et al., 2019; Dowker et al., 2016; Geary et al., 2019; Merritt, 2020). This theory of anxiety mainly pertains to women remembering the assumption that men are better at arithmetic than women, although anxiety can develop in relation to various preconceptions such as race, economic status, religion, nationality, and age (Phelps-Gregory et al., 2020; Merritt, 2020). Increased anxiety can be caused by a variety of factors, including self-confidence, exposure to stereotypes, and the influence and social transmission of anxiety by worried relatives, friends, and teachers (Phelps-Gregory et al., 2020). According to certain research, men have superior visuospatial memory than women, and math anxiety impacts verbal memory more than visuospatial working memory (DeCaro and Rittle-Johnson, 2012).

2. Nationality and Math Anxiety

Some characteristics of mathematical attitudes appear to be shared by many countries and cultures, such as the proclivity of young children to embrace mathematics. There are, however, disparities in performance, interest in mathematics, mathematical skill, and the societal value of mathematics. Children in high-performing Asian countries, such as Korea and Japan, exhibit high levels of math anxiety, whereas children in high-performing Western European countries, such as Finland, the Netherlands, Liechtenstein, and Switzerland, exhibit low levels of math anxiety (Dowker et al., 2016; Suárez-Pellicioni et al., 2016; Maloney et al., 2013; Stoet et al., 2016). There is substantial evidence that both individuals' socioeconomic condition and countries' economic situation have a significant impact on participation in mathematics programs and mathematical achievement (Dowker et al., 2016). Several studies have found that ethnic minority students in the United States and the United Kingdom had more positive attitudes toward mathematics than white students

(Aldous, 2006; Baird and Keene, 2019; Lubienski, 2002; McGraw et al., 2006; Nguyen et al., 2020).

3. Math Anxiety and Environment

Problems learning mathematics are caused by inherited factors (40-70%) and environmental variables (10-70%), according to twin studies. The heterogeneity of the groups tested explains such a wide range of values (Hart et al., 2016). According to one idea, environmental influences play a decreasing role and genetic factors play an increasing part with age; genetic factors with respect to age are as follows: Approximately 20% in infants, up to 40% in children, and 50% in teenagers (Ashcraft, 1992; Ashcraft, 2016; Ashcraft et al., 2007). Parental education level, family environment, parenting style and place of residence, technical teaching content, grading system, and teacher education level are all environmental influences (Phelps-Gregory et al., 2020).

However, socioeconomic status, particularly educational level, is a powerful predictor of early mathematics success (Dowker et al., 2016). Environmental factors such as nutrition and environmental stress, such as the presence of toxins, radiation, and pollution, among others, can also influence the function of internal structures such as the nervous system (Malanchini et al., 2017).

Educational Implications of Math Anxiety

1. Performance
2. Learning
3. Attitude Toward Math
4. Avoidance
5. Self-esteem

Moral Implications of Dyscalculia and Math Anxiety

1. Fair Evaluation

- I. **Challenge:** College students who have dyscalculia or math anxiety may have trouble with their coursework and exams that involve math.
- II. **Moral Implication:** It's critical to fairly evaluate students while taking into account their unique difficulties. Fairness principles can be upheld and academic disadvantage can be avoided by offering different assessment options or additional time.

2. Mental health matters to consider

- I. **Challenge:** Students' mental health and general wellbeing can be negatively impacted by math anxiety, which is a challenge.
- II. **Moral Implications:** It is morally required to put students' mental health first. To ensure students' emotional wellbeing, universities should provide counseling services and raise awareness about mental health issues.

3. Inclusive Education

- I. **Challenge:** It can be unethical to disallow participation to students who have dyscalculia or math anxiety because of their difficulties
- II. **Moral Implication:** Universities have a responsibility to create an inclusive environment that accommodates diverse learning needs. This promotes equal opportunities for all students and respects their dignity.

4. Supportive Resources:

- I. **Challenges:** Due to stigma, some university students might be reluctant to disclose their struggles, which could prevent them from getting the help they need.
- II. **Moral Implication:** Students with dyscalculia or math anxiety should be actively supported and given resources by universities. Open communication and the eradication of stigma can motivate students to ask for assistance without worrying about being treated unfairly.

THEORETICAL FRAMEWORK

Social Cognitive Theory

INTRODUCTION

Albert Bandura's social cognitive theory is a theoretical framework in psychology which stresses the relationship between cognition, behavior and environment. It implies that people pick up cognition not only through their own experiences but also from seeing the relationship and behavior of others. Understanding how people build up self-efficiency, motivation and behavior in numerous circumstances including learning and education is greatly impacted by this concept

Key Concept:

- 1. Observational Learning:** At the core of Social Cognitive Theory is the concept of observational learning, often referred to as modeling or vicarious learning. People learn by observing the behaviors, actions, and consequences experienced by others. This learning process enables individuals to acquire new skills and behaviors without necessarily undergoing direct experiences.
- 2. Self-Efficacy:** Self-efficacy is an individual's belief in their capacity to complete a certain task or achieve a specific goal. Bandura stated that self-efficacy is critical in determining whether people will engage in a behavior and how they will deal with problems. Due to their difficulties with mathematical tasks, students with dyscalculia may acquire reduced self-efficacy beliefs, which may impair their motivation and anxiety levels.
- 3. Outcomes Expectation:** Outcome expectations refer to an individual's anticipation of the outcomes or consequences of their actions. People are more likely to engage in behaviors that they believe will lead to positive outcomes and avoid behaviors that they believe will lead to

negative outcomes. For students with dyscalculia, the expectation of struggling or failing in mathematical tasks may contribute to heightened anxiety.

4. **Self-Regulation:** Self-regulation is emphasized in Social Cognitive Theory, which entails monitoring one's own behavior, defining objectives, and utilizing tactics to reach those goals. Students with dyscalculia may struggle to regulate their learning processes, impacting their ability to cope with mathematical activities and the anxiety that comes with it.

Application to the Study

In the context of a research project on dyscalculia and anxiety among university students, Social Cognitive Theory provides insights into how these students may acquire their attitudes, beliefs, and behaviors connected to mathematics and anxiety:

1. **Observational Learning:** Students with dyscalculia may witness peers who excel at mathematics and compare their own difficulties, thus influencing their self-efficacy views and anxiety levels.
2. **Self-Efficacy:** Understanding the function of self-efficacy will assist understand how dyscalculia-related difficulties might affect students' confidence in completing mathematical activities and their ability to manage accompanying anxiety.
3. **Outcome Expectations:** The theory's emphasis on outcome expectations explains how students' expectations about their mathematical achievement might influence their emotional responses, including fear.
4. **Self-Regulation:** Social Cognitive Theory emphasizes the role of self-regulation in learning and anxiety management. Students with dyscalculia may have difficulty developing appropriate self-regulation skills, resulting in heightened anxiety during mathematical assignments.

Conclusion: Social Cognitive Theory provides a comprehensive framework for understanding how individuals, including university students with dyscalculia, learn, develop beliefs, and engage in behaviors related to mathematics and anxiety. By taking into account the reciprocal interactions between cognitive processes, behaviors, and the environment, the theory sheds light on the potential mechanisms underlying the relationship between dyscalculia and anxiety levels in the context of higher education.

Cognitive Load Theory

INTRODUCTION

Cognitive Load Theory (CLT) is a concept in educational psychology that centers on the cognitive effort needed for learning as well as the limitations of working memory. CLT, which was developed in the 1980s by John Sweller, tries to explain how the cognitive load placed on learners by instructional materials affects learning results. It offers perspectives on instructional design and how to enhance learning by properly regulating cognitive load.

Key Concepts:

1. **Working Memory:** The concept of working memory, which is a temporary storage system for information processing, is central to CLT. Working memory has limited storage capacity, which implies it is able to retain a little piece of information at a time and for a short period of time. According to cognitive load theory, individuals have limitations on the quantity of data they are able to process concurrently in working memory.
2. **Cognitive Load:** The mental effort required to process information is referred to as cognitive load. CLT differentiates three forms of cognitive load:

- I. **Intrinsic Cognitive Load:** This is the inherent difficulty of a particular undertaking.
Intrinsic cognitive load occurs in tasks that are more complicated or demand more mental processing.
- II. **Extraneous Cognitive Load:** The mental effort exerted by the instructional materials or design is referred to as extraneous cognitive load. Unsuitable materials might increase additional cognitive strain and impede learning.
- III. **Germane Cognitive Load:** This is the cognitive effort expended in comprehending and learning the subject. It is said to be advantageous since it aids in understanding as well as learning.

Application to Learning

1. **Reduce Extraneous Load:** According to CLT, instructional materials and design should attempt to reduce superfluous cognitive load. Clear and well-structured presentations, organized information, and user-friendly interfaces can assist learners in focusing on the content rather than the presentation.
2. **Manage Intrinsic Load:** While intrinsic load is inherent in the subject matter's complexity, CLT calls for sequencing and scaffolding. To alleviate cognitive overload, complex concepts can be broken down into smaller, more manageable chunks.
3. **Optimize Germane Load:** CLT advises that learners' cognitive efforts be directed toward comprehension and problem-solving (germane load) to facilitate learning. This can be accomplished through efficient exercises, interactive activities, and opportunities for further material engagement.

Implications for Education

- 1. Effective Instructional Design:** CLT promotes educators to create learning materials and activities that are appropriate for the cognitive abilities of their students. This includes clear instructions, the use of suitable multimedia, and chances for practice and reflection.
- 2. Spaced Learning:** CLT advocates for the concept of spaced or distributed learning. rather than instead of cramming information, students gain from reviewing content over time, which allows for greater encoding into long-term memory and reduces cognitive burden during each session.
- 3. Scaffolded Learning:** CLT principles are aligned with progressive disclosure of information and scaffolding (offering support as learners develop). This allows students to gain knowledge without feeling overwhelmed.

Conclusion: Cognitive Load Theory provides useful insights into how instructional design and delivery affect learning results. Understanding the cognitive load imposed on learners allows educators to tailor their teaching techniques and resources to improve information comprehension and retention. This notion is especially pertinent in today's educational environment, as technology and multimedia play an important role in instructional delivery.\

Mathematics Anxiety Theory

INTRODUCTION

The emotional and psychological elements of studying and performing mathematics are the subject of Mathematics Anxiety Theory. It investigates how people's feelings and attitudes toward mathematics influence their cognitive and behavioral responses to mathematical activities. This

hypothesis accepts that arithmetic anxiety is a real and pervasive problem that affects people of all ages and educational levels.

Key Concepts

- 1. Mathematics Anxiety:** Mathematics anxiety is a sort of anxiety that is characterized by worry or uneasiness when performing mathematical tasks such as problem solving or calculations. Anxiety symptoms might include both physical (e.g., sweating) and cognitive (e.g., worried about making mistakes).
- 2. Emotional Responses:** According to Mathematics Anxiety Theory, anxiety is an emotional response to mathematics that can affect performance. Anxiety about mathematics might affect a person's ability to think clearly and solve mathematical issues successfully.
- 3. Cognitive Interference:** According to the hypothesis, mathematics anxiety can cause cognitive interference, in which anxious thoughts and fears disturb the cognitive processes required for mathematical activities. This interference can lead to impaired working memory capacity and problem-solving ability.
- 4. Self-Perception:** Individuals with mathematics anxiety frequently have negative self-perceptions about their mathematical abilities. These unfavorable self-perceptions can amplify anxiety and lead to a self-fulfilling cycle of low performance.

Factors Contributing to Mathematics Anxiety

- 1. Previous Experiences:** Previous negative mathematical experiences, such as failures or criticism, might contribute to mathematics anxiety. Individuals may link mathematics with stress and failure as a result of these experiences.

2. **Social and Cultural Factors:** An individual's perspective of mathematics can be influenced by societal attitudes and cultural beliefs about the topic. A culture that values mathematical ability may put more pressure on individuals, raising the incidence of mathematics anxiety.
3. **Instructional Methods:** Mathematics anxiety can be exacerbated by ineffective or anxiety-inducing instructional approaches. Teachers who prioritize rote memorization over comprehension or who create a high-pressure climate in the classroom can worsen anxiety.

Implications and Interventions

1. **Supportive Learning Environments:** Developing an encouraging and nonjudgmental classroom or learning atmosphere might help alleviate mathematical anxiety. It can be useful to encourage a growth mentality, in which mistakes are viewed as opportunities for learning.
2. **Effective Pedagogy:** Anxiety can be reduced by effective pedagogy, which emphasizes conceptual understanding and problem solving over memorization. Clear explanations and guidance might also be beneficial.
3. **Gradual Exposure:** Increasing the difficulty of mathematics activities gradually can help individuals acquire confidence and minimize fear over time.
4. **Counseling and Psychological Support:** Individuals suffering from severe mathematics anxiety may benefit from counseling and psychological support. Cognitive-behavioral therapy (CBT) is one technique that can help people control their anxiety responses.

Conclusion: The Mathematics Anxiety Theory emphasizes the significance of understanding and resolving the emotional and psychological aspects of mathematics learning. Educators and individuals can attempt to alleviate mathematics anxiety and promote more positive attitudes and

outcomes in mathematics education by recognizing the variables that contribute to it and adopting effective treatments.

SUMMARY

1. Dyscalculia in University Students:

- I. Several studies have highlighted the presence of dyscalculia, a specific learning disability in mathematics, among university students across various disciplines.
- II. Dyscalculia is associated with persistent difficulties in understanding and manipulating numerical concepts, which can lead to academic challenges and reduced self-esteem.

2. Anxiety Levels in University Students:

- I. Numerous research works have emphasized the prevalence of anxiety among university students, including generalized anxiety and discipline-specific anxieties, such as mathematics anxiety.
- II. Anxiety in university students has been linked to academic performance, self-confidence, and overall well-being.

3. Correlation between Dyscalculia and Anxiety:

- I. While there is a significant body of literature examining dyscalculia and anxiety separately, limited research has explored the relationship between these two factors in university students.
- II. Some preliminary studies suggest a potential correlation, with students with dyscalculia reporting higher levels of mathematics anxiety.

4. Psychosocial Implications:

- I. Research has indicated that dyscalculia can have psychosocial implications, including increased stress and anxiety, due to the challenges associated with mathematics.

- II. Anxiety related to dyscalculia can further hinder academic performance and contribute to a negative cycle of math-related stress.

5. Educational Interventions:

- I. Literature highlights the importance of implementing educational interventions and support mechanisms for students with dyscalculia, including specialized instruction, tutoring, and accommodation.
- II. Effective interventions can lead to improved mathematical skills and potentially reduced anxiety levels.

6. Theoretical Frameworks:

- I. The literature review also identified relevant theoretical frameworks, such as Social Cognitive Theory, Cognitive Load Theory, and Mathematics Anxiety Theory, which can provide insights into the psychological and cognitive aspects of dyscalculia and anxiety in university students.

7. Research Gap and Rationale:

- I. The literature review underscores the need for further research to systematically investigate the relationship between dyscalculia and anxiety levels among university students, as this intersection remains relatively unexplored.
- II. By addressing this research gap, the proposed study aims to contribute to a deeper understanding of the challenges faced by university students and inform potential interventions to support their academic success and emotional well-being.

CHAPTER THREE

RESEARCH METHODOLOGY

Introduction

This chapter described the procedure used to gather data for the research project. The study was designed to find out the relationship between dyscalculia and math anxiety level among university students in Nigeria. The method used to carry out this study under the following subheading

Research Design

Population of the study

Sample and sampling technique

Research instrument

Validity of instruments

Reliability of instruments

Method of Data Collection

Method of Data Analysis

Research Design

The strategy employed in this research is a cross-sectional research design.

Population of the Study

For this study, participants across 6 different universities in Nigeria were selected. The total population of study is 301 students

Research Instrument

The Mathematics Anxiety Assessment Questionnaire (MAAQ) was drawn on a 5-point scale of: Strongly agree (SA), Agreed (A), Neutral (N), Disagree (D) and Agree (A). The Dyscalculia Assessment Questionnaire (DAQ) was drawn on a 5-point scale of: Always, Often, Sometimes, Occasionally, and Not At All. The scales have been assigned values of 5,4,3,2, and 1 respectively. The respondents were to respond by ticking the alternative provided below each statement.

Validation of the Instrument

The researchers' supervisor and a lecturer in the department of mathematics validated the questionnaire. This is done to ensure that the questions asked are appropriate and in line with the research variables for the study. Thirty-seven questions were presented to the supervisor and lecturer by the researcher. The suggestions and correlations made by the supervisor were further incorporated in the final draft of the questionnaire

Method of Data Analysis

1. Frequency counts and charts were used to summarize demographic characteristics of the sample.
2. Regression analysis was used to examine the relationship between dyscalculia and anxiety levels.

CHAPTER FOUR

PRESENTATION OF RESULTS AND DISCUSSIONS OF FINDINGS

The results of this study are carefully represented in this chapter according to the research questions and hypotheses.

The Mathematics Anxiety Assessment Questionnaire (MAAQ) and the Dyscalculia Assessment questionnaire (DAQ) were administered to students across six universities in Nigeria. The data collected was analyzed to answer the research questions.

DEMOGRAPHIC FACTORS

CHART 1

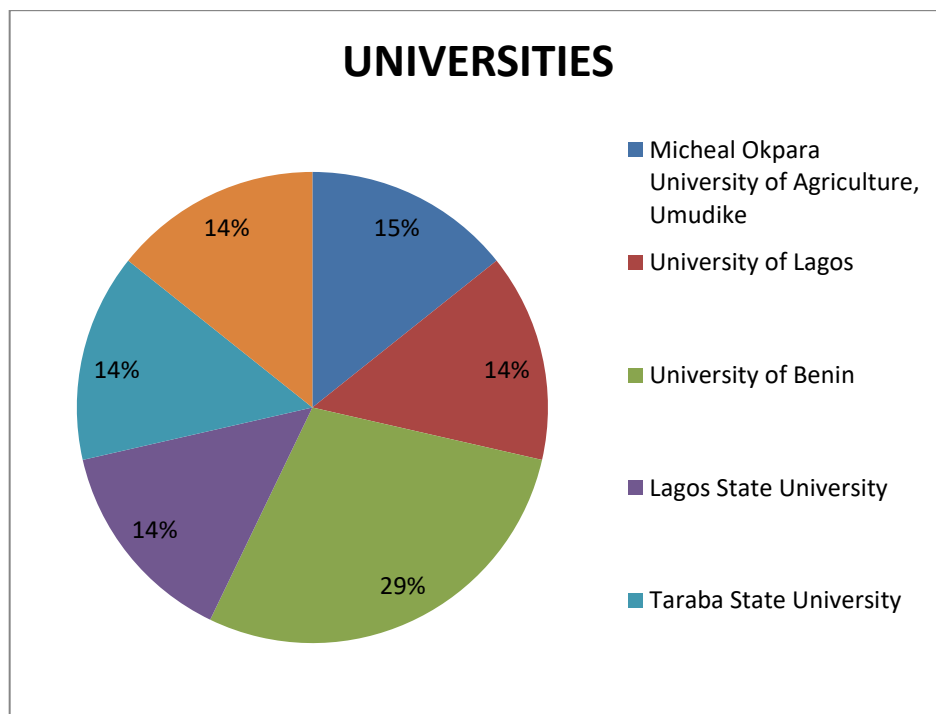


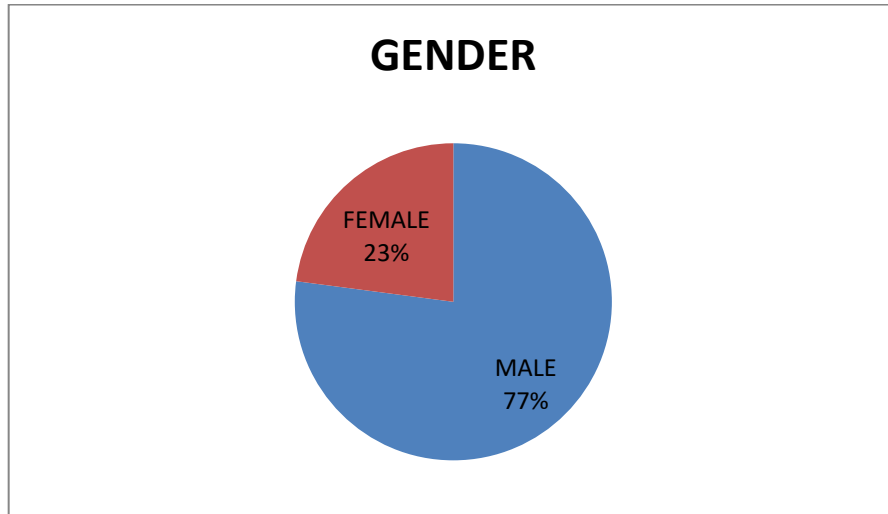
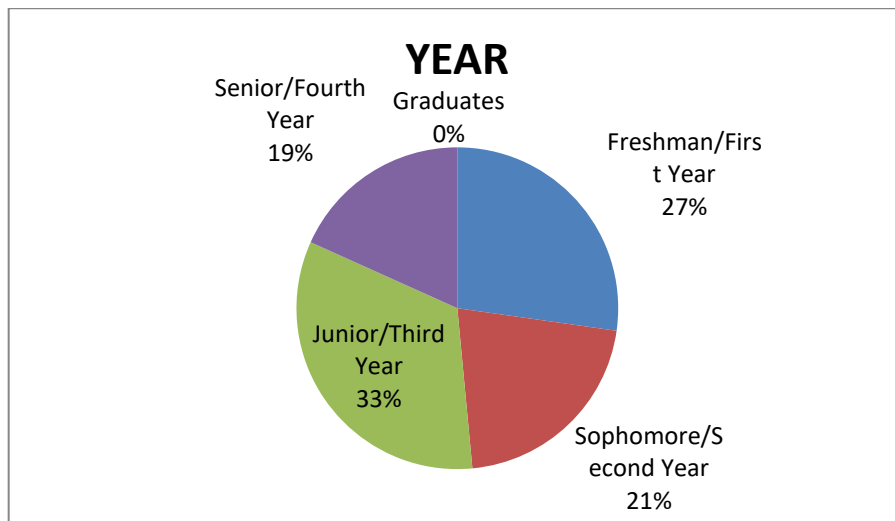
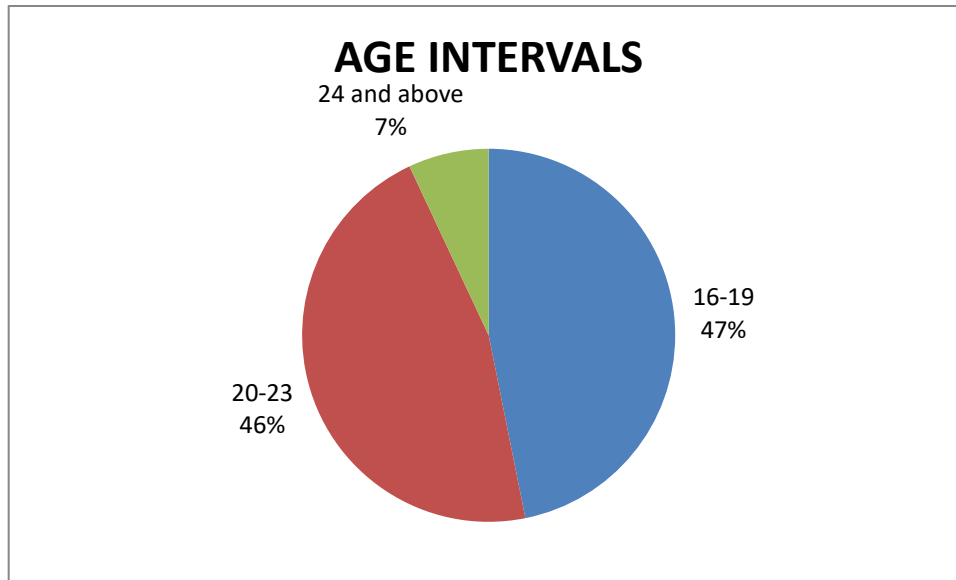
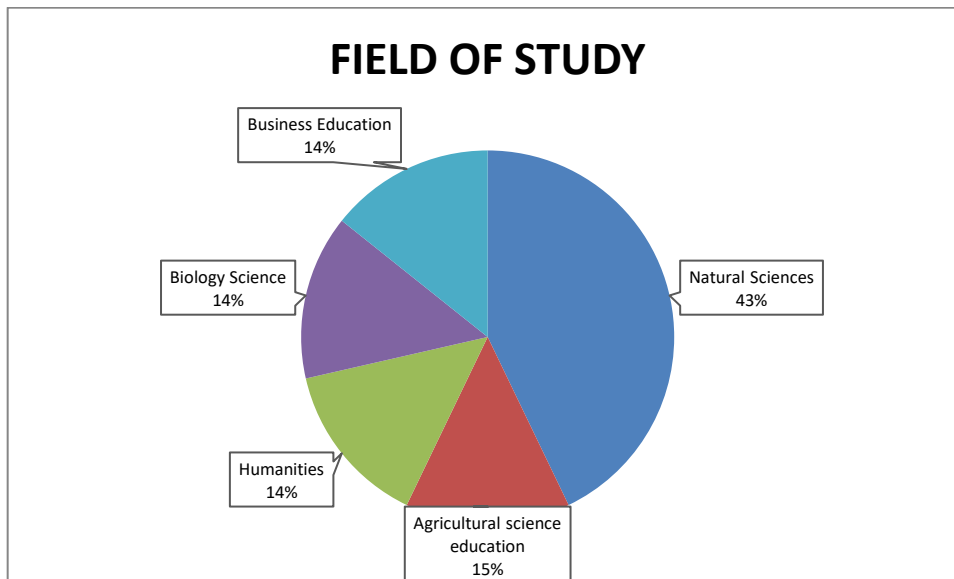
CHART 2**CHART 3**

CHART 4**CHART 5**

Research Question One

1. Is there a statistically significant correlation between the presence of dyscalculia and anxiety levels among university students?

Below is the result of the regression statistics carried out on 301 students' responses by the researcher showing correlation between the presence of dyscalculia and anxiety levels among university students.

TABLE 1:

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.796418
R Square	0.634282
Adjusted R Square	0.633059
Standard Error	7.81337
Observations	301

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	31658	31658	518.5692	2.83E-67
Residual	299	18253.58	61.04875		
Total	300	49911.58			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	71.12688	1.488722	47.77715	5.3E-142	68.19718	74.05658
Presence of Dyscalculia	-0.63757	0.027998	-22.7721	2.83E-67	-0.69267	-0.58248

The Multiple R is approximately 0.796, indicating a moderate to strong positive correlation between these variables.

An R Square of 0.634 means that about 63.4% of the variability in "Mathematics Anxiety" can be explained by "Presence of Dyscalculia."

The regression sum of squares (SS) is 49911.58. The very low p-value (2.83266E-67) indicates that the model is statistically significant.

The coefficient (-0.638) for "Presence of Dyscalculia" represents the change in the predicted value of "Mathematics Anxiety" for a one-unit change in "Presence of Dyscalculia." The t-statistic (-22.772) and very low p-value (2.83266E-67) indicate that this coefficient is statistically significant.

The 95% confidence interval suggests that the true coefficient falls between -0.693 and -0.582.

Research Hypothesis One

There is a strong negative relationship between "Presence of Dyscalculia" and "Mathematics Anxiety". As "Presence of Dyscalculia" increases (indicating the absence of dyscalculia), the predicted level of "Mathematics Anxiety" decreases. In simpler terms, students without dyscalculia tend to have lower levels of mathematics anxiety.

In summary, this analysis suggests that the presence of dyscalculia is a significant predictor of lower mathematics anxiety, with students without dyscalculia generally having lower levels of mathematics anxiety. The model provides a statistically significant explanation for the variability in mathematics anxiety.

Research Question Two

To what extent do different aspects of dyscalculia such as numerical processing, mathematical reasoning, and arithmetic skills correlate with varying levels of anxiety in University students?

Below are the results of the regression statistics carried out on 301 students' responses by the researcher showing correlation between the different aspects of dyscalculia and anxiety levels among university students.

Table 2:

This shows the regression statistics and correlation between Number Sense and Mathematics Anxiety level.

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.777029
R Square	0.603775
Adjusted R Square	0.60245
Standard Error	8.132726
Observations	301

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	30135.35	30135.35	455.6213	4.62E-62
Residual	299	19776.23	66.14123		
Total	300	49911.58			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>
Intercept	69.36458	1.506068	46.05675	8.3E-138	66.40075	72.32842	66.40075
NUMBER SENSE	-2.11154	0.098923	-21.3453	4.62E-62	-2.30621	-1.91686	-2.30621

The Multiple R is the correlation coefficient between the independent variable ("NUMBER SENSE") and the dependent variable ("Mathematics Anxiety Level"). It's approximately 0.778, indicating a moderate to strong positive correlation between "NUMBER SENSE" and "Mathematics Anxiety Level."

An R Square of 0.604 means that about 60.4% of the variability in "Mathematics Anxiety Level" can be explained by "NUMBER SENSE."

The regression sum of squares (SS) is 30135.35. The very low p-value (4.62076E-62) indicates that it is statistically significant.

The intercept represents the predicted value of "Mathematics Anxiety Level" when "NUMBER SENSE" is zero. It's approximately 68.462. The coefficient (-2.112) for "NUMBER SENSE" represents the change in the predicted value of "Mathematics Anxiety Level" for a one-unit change in "NUMBER SENSE." The t-statistic (-21.345) and very low p-value (4.6208E-62) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.306 and -1.917.

The regression analysis suggests that there is a strong negative relationship between "NUMBER SENSE" and mathematics anxiety level. As "NUMBER SENSE" improves (increases), the predicted value of the dependent variable decreases. In other words, better number sense skills are associated with lower values of the mathematics anxiety level.

Overall, this analysis suggests that "NUMBER SENSE" is a significant predictor of "Mathematics Anxiety Level," with higher "NUMBER SENSE" scores associated with lower levels of mathematics anxiety.

Table 3

This table shows the regression analysis between Mathematics Anxiety and Working Memory

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.746209
R Square	0.556828
Adjusted R Square	0.555346
Standard Error	8.601044
Observations	301

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	27792.17	27792.17	375.6818	8.96E-55

Residual	299	22119.41	73.97795
Total	300	49911.58	

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	67.18722	1.545528	43.47201	2.8E-131	64.14573	70.22872
WORKING MEMORY	-2.45272	0.126543	-19.3825	8.96E-55	-2.70174	-2.20369

The Multiple R is the correlation coefficient between "WORKING MEMORY" and the dependent variable (Mathematics anxiety Level). It's approximately 0.746, indicating a moderate to strong positive correlation between these variables.

An R Square of 0.557 means that about 55.7% of the variability in Mathematics Anxiety Level can be explained by "WORKING MEMORY."

The regression sum of squares (SS) is 27792.17. The F-statistic tests whether the regression model is significant. The very low p-value (8.95942E-55) indicates that the model is statistically significant.

The coefficient (-2.453) for "WORKING MEMORY" represents the change in the predicted value of Mathematics Anxiety Level for a one-unit change in "WORKING MEMORY." The t-statistic (-19.383) and very low p-value (8.95942E-55) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.702 and -2.204.

There is a strong negative relationship between "WORKING MEMORY" and mathematics anxiety level. As "WORKING MEMORY" increases (improves), the predicted value of the dependent variable decreases. In other words, better working memory is associated with lower values of the dependent variable.

In summary, this analysis suggests that "WORKING MEMORY" is also a significant predictor of the dependent variable, with better working memory being associated with lower values of the dependent variable. The model provides a statistically significant explanation for the variability in the dependent variable.

Table 4:

The table below shows the analysis between Mathematics Anxiety Level and Accuracy and Fluent Calculation

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.778749
R Square	0.60645
Adjusted R Square	0.605134
Standard Error	8.105225
Observations	301

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	30268.87	30268.87	460.7508	1.67E-62
Residual	299	19642.71	65.69467		
Total	300	49911.58			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	69.21852	1.491518	46.40809	1.1E-138	66.28332	72.15372
Accuracy and Fluent Calculation	-2.43593	0.113483	-21.4651	1.67E-62	-2.65925	-2.2126

The Multiple R is approximately 0.779, indicating a moderate to strong positive correlation between these variables.

An R Square of 0.606 means that about 60.6% of the variability in Mathematics Anxiety Level can be explained by "Accuracy and Fluent Calculation."

The regression sum of squares (SS) is 30268.87. The very low p-value (1.67461E-62) indicates that the model is statistically significant.

The coefficient (-2.436) for "Accuracy and Fluent Calculation" represents the change in the predicted value of the dependent variable for a one-unit change in "Accuracy and Fluent Calculation." The t-statistic (-21.465) and very low p-value (1.67461E-62) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.659 and -2.213.

The analysis suggests that there is a strong negative relationship between "Accuracy and Fluent Calculation" and Mathematics Anxiety Level. As "Accuracy and Fluent Calculation" improves (increases), the predicted value of Mathematics Anxiety Level decreases. In other words, better accuracy and fluency in mathematical calculations are associated with lower values of the mathematics anxiety level.

In summary, this analysis suggests that "Accuracy and Fluent Calculation" is also a significant predictor of mathematics anxiety level, with better accuracy and fluency in mathematical calculations being associated with lower values of mathematics anxiety level.

Table 5:

The table below shows the analysis between mathematics anxiety level and mathematics reasoning

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.780285
R Square	0.608845

Adjusted R Square	0.607536
Standard Error	8.080527
Observations	301

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	30388.4	30388.4	465.4022	6.71E-63
Residual	299	19523.18	65.29492		
Total	300	49911.58			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
Intercept	70.60849	1.545642	45.6823	7E-137	67.56677	73.6502
MATHEMATICS REASONING	-2.61408	0.121173	-21.5732	6.71E-63	-2.85254	-2.37562

The Multiple R is approximately 0.780, indicating a moderate to strong positive correlation between these variables.

An R Square of 0.609 means that about 60.9% of the variability in mathematics anxiety level can be explained by "MATHEMATICS REASONING."

The regression sum of squares (SS) is 30388.4. The F-statistic tests whether the regression model is significant. The very low p-value (6.71089E-63) indicates that the model is statistically significant.

The coefficient (-2.614) for "MATHEMATICS REASONING" represents the change in the predicted value of mathematics anxiety level for a one-unit change in "MATHEMATICS REASONING." The t-statistics (-21.573) and very low p-value (6.71089E-63) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.853 and -2.376.

The analysis suggests that there is a strong negative relationship between "MATHEMATICS REASONING" and the Mathematics Anxiety Level. As "MATHEMATICS REASONING" improves (increases), the predicted value of the mathematics anxiety level decreases. In other words, better mathematics reasoning skills are associated with lower values of mathematics anxiety level.

In summary, this analysis suggests that "MATHEMATICS REASONING" is a significant predictor of the mathematics anxiety level in university students in Nigeria, with better mathematics reasoning skills being associated with lower values of the mathematics anxiety level.

Hypothesis Two:

There is a strong negative correlation between the various aspects of dyscalculia and anxiety levels in university students in Nigeria. Higher performance in various mathematical aspects indicates a lower anxiety level.

CHAPTER FIVE

SUMMARY OF FINDINGS AND RECOMMENDATION

From Chart 1, six schools were used for this research, with a total of 301 respondents. 43(14.29%) respondents were from Micheal Okpara University of Education, Umudike, 43 (14.29%) respondents were from University of Lagos, 86 (29%) were from the University of Benin, Benin city, 43(14.29%) respondents were from Tai Solarin State University of education, 43(14.29%) respondents were from Lagos State University and 43(14.29%) were from Taraba State University.

Chart 2 showed that 232(73%) Males and 69(27%) Females were used for this research.

Chart 3 showed that 23% were Year 1 students, 21% were Year 2 students, 33% were Year 3 students, 19% were Year 4 students and 0% was Graduate students

Chart 4 showed the age range of students used in the research. 47% of students ranged from 16-19 years, 46% of students ranged from 20-23 and 7% of students ranged from 24 and above.

Chart 5 showed the field of study of the respondents. It showed that 43% of the population were from the Natural Science, 14.29% were from the agricultural education, 14.29% of the students were from the humanities department, 14.29% of the respondents were from the biology science and 14.29% of the population were from the Business education department.

Table 1 showed the regression analysis between mathematics anxiety level in university students in Nigeria and Dyscalculia levels. From the analysis, it was seen that there was a strong negative correlation of -0.638 between dyscalculia levels in university students and mathematics anxiety level. As the dyscalculic state increases, the mathematics anxiety level also increases.

Table 2 showed the regression analysis between mathematics anxiety level and number sense. The regression analysis suggests that there is a strong negative relationship between "NUMBER

SENSE" and mathematics anxiety level. As "NUMBER SENSE" improves (increases), the predicted value of the dependent variable decreases. In other words, better number sense skills are associated with lower values of the mathematics anxiety level.

Table 3 showed the regression analysis between mathematics anxiety level and working memory. The coefficient (-2.453) for "WORKING MEMORY" represents the change in the predicted value of Mathematics Anxiety Level for a one-unit change in "WORKING MEMORY." The t-statistics (-19.383) and very low p-value (8.95942E-55) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.702 and -2.204. There is a strong negative relationship between "WORKING MEMORY" and mathematics anxiety level. As "WORKING MEMORY" increases (improves), the predicted value of the dependent variable decreases. In other words, better working memory is associated with lower values of the dependent variable.

Table 4 showed the regression analysis between Mathematics anxiety level and Accuracy and Fluent Calculation. The coefficient (-2.436) for "Accuracy and Fluent Calculation" represents the change in the predicted value of the dependent variable for a one-unit change in "Accuracy and Fluent Calculation." The t-statistics (-21.465) and very low p-value (1.67461E-62) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.659 and -2.213. The analysis suggests that there is a strong negative relationship between "Accuracy and Fluent Calculation" and Mathematics Anxiety Level. As "Accuracy and Fluent Calculation" improves (increases), the predicted value of Mathematics Anxiety Level decreases. In other words, better accuracy and fluency in mathematical calculations are associated with lower values of the mathematics anxiety level.

Table 5 shows the regression analysis between Mathematics anxiety level and mathematics reasoning. The coefficient (-2.614) for "MATHEMATICS REASONING" represents the change in the predicted value of mathematics anxiety level for a one-unit change in "MATHEMATICS REASONING." The t-statistics (-21.573) and very low p-value (6.71089E-63) indicate that this coefficient is statistically significant. The 95% confidence interval suggests that the true coefficient falls between -2.853 and -2.376.

The analysis suggests that there is a strong negative relationship between "MATHEMATICS REASONING" and the Mathematics Anxiety Level. As "MATHEMATICS REASONING" improves (increases), the predicted value of the mathematics anxiety level decreases. In other words, better mathematics reasoning skills are associated with lower values of mathematics anxiety level.

CONCLUSION

It can be said that there is a strong negative correlation between mathematics anxiety level and dyscalculia in university students in Nigeria. It shows that a rise in dyscalculic state of students has resulted to an increase in mathematics anxiety level of the students. It can also be seen that there is a strong negative correlation between the factors of dyscalculia and mathematics anxiety level among university students in Nigeria.

RECOMMENDATIONS

1. Educators and Teachers:

Educators should be made aware of the strong negative correlation between dyscalculia and anxiety levels among university students.

Also, training programs for educators could include strategies to identify students with dyscalculia early and implement supportive teaching methods to reduce anxiety in mathematics-related courses.

Providing resources and professional development opportunities for educators to address the emotional well-being of students with dyscalculia is essential.

2. University Administrators:

University administrators should consider implementing support services and accommodations for students with dyscalculia to help alleviate their anxiety.

Also promoting a more inclusive and accommodating learning environment can improve the overall academic experience for students with dyscalculia.

3. Psychologists and Counselors:

Mental health professionals should be informed about the link between dyscalculia and anxiety levels.

Also, they can provide targeted counseling and interventions to university students with dyscalculia who may experience heightened anxiety related to their mathematical difficulties.

4. Students with Dyscalculia:

Students with dyscalculia should be informed about the potential for increased anxiety related to mathematics.

Also, they should be encouraged to seek support services available at the university, such as tutoring, counseling, or accommodations, to help them manage both their mathematical challenges and anxiety.

5. Parents and Caregivers:

Parents and caregivers of university students with dyscalculia should be aware of the potential emotional challenges their children may face.

Also, encourage open communication with their children and support their access to appropriate resources and services at the university.

6. Society at Large:

Greater awareness of the negative correlation between dyscalculia and anxiety in university students can contribute to a more understanding and supportive society.

Also, reducing stigma and fostering inclusivity in education and workplaces can benefit individuals with dyscalculia.

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